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2.4 Membrane Proteins

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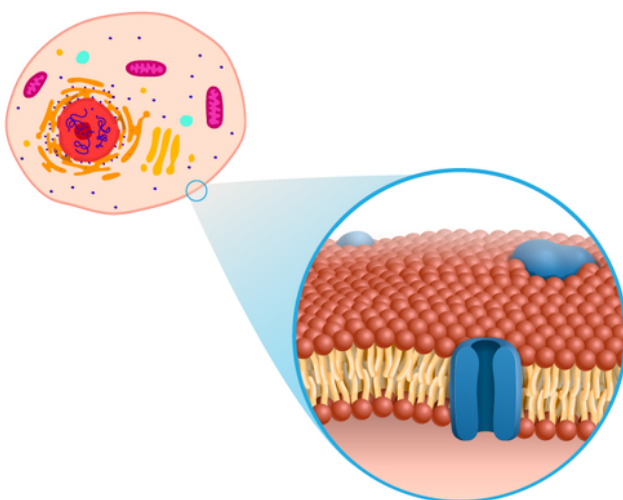
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What will you learn

- Types of membrane proteins
- The **Fluid Mosaic Model** of the cell membrane
- Different types of extensions of the cell membrane and their role in cell movement



[Figure 1]

Can anything or everything move in or out of the cell?

No. It is the **semipermeable** cell membrane that determines what can enter and leave the cell. So, if not everything can cross the membrane, how do certain things get across?

Molecules of cholesterol help the cell membrane keep its shape. Many of the proteins in the cell membrane assist other substances in crossing the membrane.

The cell membranes also contain certain types of proteins. A membrane protein is a protein molecule that is attached to, or associated with, the membrane of a cell or an **organelle**. Membrane proteins can be put into two groups based on how the protein is associated with the membrane.

Integral membrane proteins are permanently embedded within the cell membrane. They have a range of important functions. Such functions include channeling or transporting molecules across the membrane. Other integral proteins act as cell receptors. Integral membrane proteins can be classified according to their relationship with the bilayer:

- Transmembrane proteins span the entire cell membrane. Transmembrane proteins are found in all types of biological membranes.
- Integral monotopic proteins are permanently attached to the membrane from only one side.

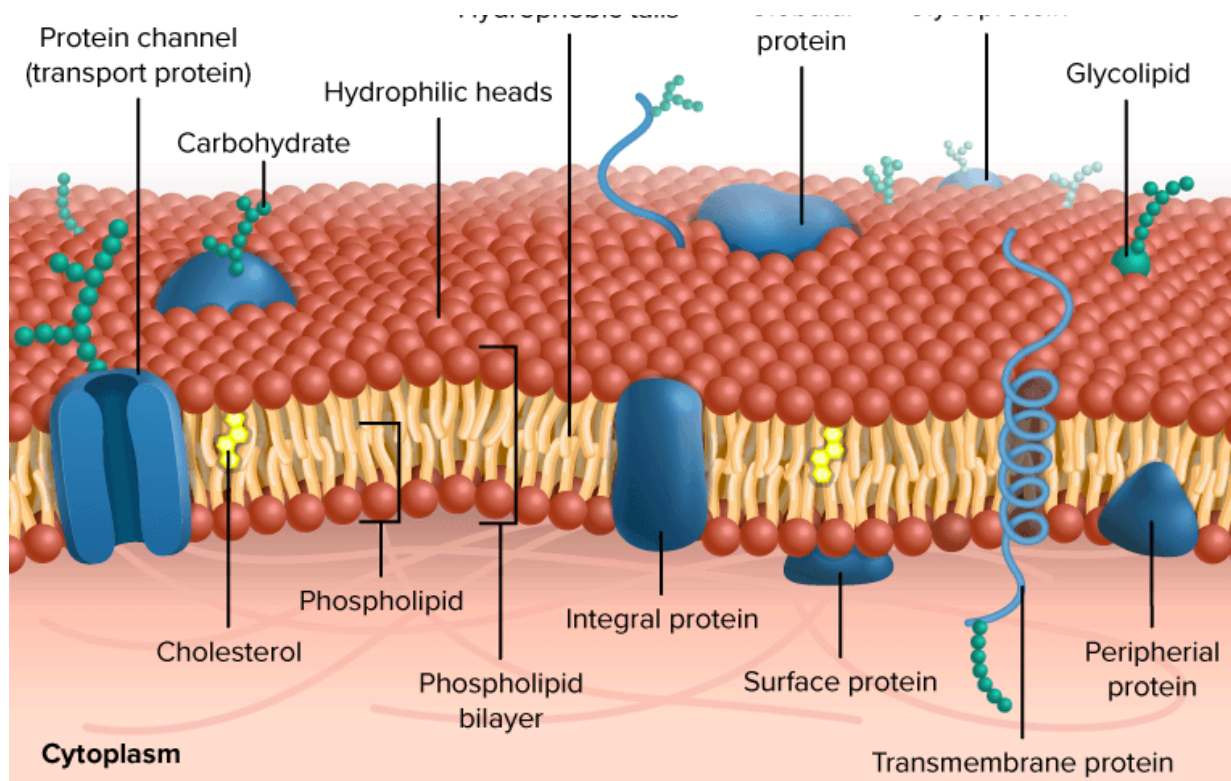
Some integral membrane proteins are responsible for cell **adhesion** (sticking of a cell to another cell or surface). On the outside of cell membranes and attached to some of the proteins are **carbohydrate** chains that act as labels that identify the cell type. Shown in the **Figure below** are two different types of membrane proteins and associated molecules.



DID YOU KNOW?

Glycoproteins and glycolipids have a carbohydrate chain that acts as a label to identify the cell type. Specifically, A, B, O **blood** groups result from having different carbohydrate chains on the cell surface of **red blood cells**.

Peripheral membrane proteins are proteins that are only temporarily associated with the membrane. They can be easily removed, which allows them to be involved in cell signaling. Peripheral proteins can also be attached to integral membrane proteins, or they can stick into a small portion of the lipid bilayer by themselves. Peripheral membrane proteins are often associated with **ion channels** and transmembrane receptors. Most peripheral membrane proteins are **hydrophilic**.



[Figure 2]

Some of the membrane proteins make up a major transport system that moves molecules and ions through the polar phospholipid bilayer.

DID YOU KNOW?

Phospholipase is an **enzyme** that breaks down various bonds in the polar head group of phospholipids. These enzymes are important to help break down damaged or aged cell membranes. This enzyme represents an example of a peripheral membrane protein.

The Fluid Mosaic Model

In 1972, S.J. Singer and G.L. Nicolson proposed the now widely accepted **Fluid Mosaic Model** of the structure of cell membranes. The **model** proposes that integral membrane proteins are embedded in the **phospholipid bilayer**, as seen in the **Figure above**. Some of these proteins extend all the way through the bilayer, and some only partially across it. These membrane proteins act as **transport proteins** and receptors proteins.

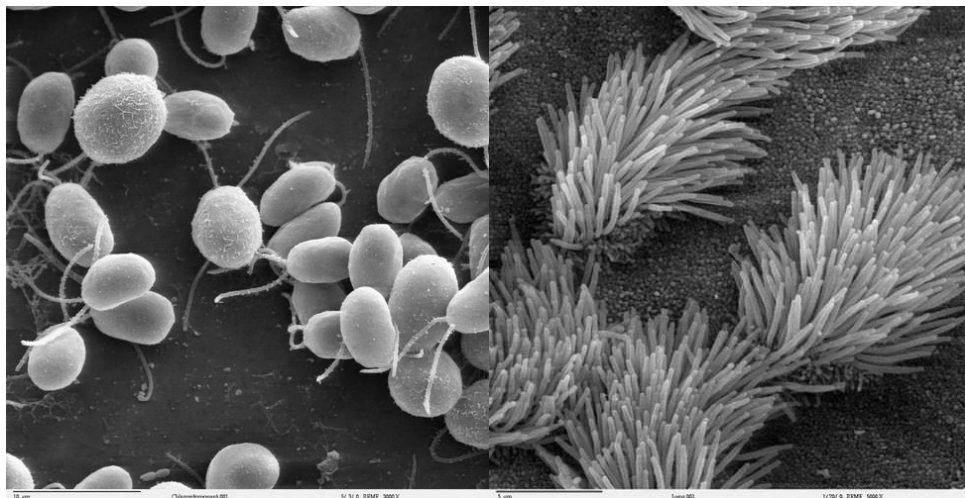
Their model also proposed that the membrane behaves like a fluid, rather than a solid. The proteins and lipids of the membrane move around the membrane, much like buoys in **water**. Such movement causes a constant change in the "mosaic pattern" of the cell membrane.





Extensions of the Cell Membrane

The cell membrane may have extensions, such as whip-like **flagella** or brush-like **cilia**. In single-celled **organisms**, like those shown in the **Figure below**, the membrane extensions may help the organisms move. In **multicellular organisms**, the extensions have other functions. For example, the cilia on human **lung** cells sweep foreign particles and **mucus** toward the mouth and nose.



[Figure 3]

[Figure 4]

Flagella in Chlamydomonas and Cilia in Bronchiolar epithelium. Flagella and Cilia are extensions of the cell membrane of many cells.



- The Fluid Mosaic Model depicts the biological nature of the cell membrane.
- Cilia and flagella are extensions of the cell membrane.

Review

1. What is the main difference between the two main types of proteins associated with the cell membrane?
2. What are two functions of integral membrane proteins?
3. Discuss the Fluid Mosaic Model.
4. What are flagella and cilia?



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